

AWS/UNIX 2026

Material by Jose Hector Galvez, Zhibin Lu & Rob Syme

Contents

Chapter 1

AWS/UNIX primer info

The UNIX exercises here are modified versions of the Software Carpentry Unix shell lesson. Licensed under CC-BY 4.0 2018–2021 by The Carpentries.

1.0.1 Data download

We will be using Amazon Web Services for the primer session. If you are not able to attend this session live, you can download the data we used here:

- data.zip
- gff3

You will need a UNIX environment (like a Linux/Mac computer or Windows Subsystem for Linux) to work through the material.

1.0.2 Schedule

RNA-seq Analysis AWS/UNIX pre-session 2026: Tuesday, June 16, 2026

(ET)	Module
11:00	Welcome Introduction
11:10	Module 1: Logging into AWS
12:10	Short Break
12:15	Module 2: Introduction to the UNIX Command Line
12:35	Module 3: File Manipulation
13:05	Long Break
14:05	Module 4: Searching and Sorting Through Files

(ET)	Module
15:00	Short Break
15:05	Module 5: Shell scripts
16:00	Finished

Chapter 2

Meet Your Faculty

2.0.1 Instructors

2.0.1.1 Diogo Pellegrina (he/him)

Postdoctoral Fellow University of Saskatchewan, VIDO Saskatoon,
SK, Canada

Diogo is a Postdoctoral Fellow at the Vaccines and Infectious Diseases Organisation (VIDO) at USask. Previously I did my PhD in the University of Sao Paulo, and I was also a Postdoc at OICR. I worked with many types of Bioinformatics pipelines, but mostly with Transcriptomics and Network Analysis.

2.0.1.2 Zhibin Lu (he/his)

Senior Manager, Digital Research University Health Network
Toronto, ON, Canada

— zhibin@gmail.com

Zhibin Lu is a senior manager at University Health Network Digital. He is responsible for UHN HPC operations and scientific software. He manages two HPC clusters at UHN, including system administration, user management, and maintenance of bioinformatics tools for HPC4health. He is also skilled in Next-Gen sequence data analysis and has developed and maintained bioinformatics pipelines at the Bioinformatics and HPC Core. He is a member of the Digital Research Alliance of Canada Bioinformatics National Team and Scheduling National Team.

2.0.2 Teaching Assistants

2.0.2.1 Alyona Ivanova

PhD Candidate University of Toronto, The Hospital for Sick Children Toronto, BC, Canada

— alyona.ivanova@mail.utoronto.ca

Alyona is a graduate student at the University of Toronto pursuing PhD in Medical Sciences. Her research focuses on identifying novel therapies for targeting chemo-resistance in glioblastoma using spatial 'omics technologies. She works in close collaboration with clinicians, neuropathologists, scientists bridging the gap between clinical practice and fundamental lab research. Alyona is a Creative Director of Panoramics - A Vision, and an Executive Editor and Director of Distribution of the Institute of Medical Sciences Magazine.

2.0.3 Facilitator

2.0.3.1 Neha Ratti (she/her)

Regional Coordinator, CBH Ontario Canadian Bioinformatics Hub, University Health Network Toronto, ON, Canada

— ontario@bioinformatics.ca

Neha is passionate about advancing bioinformatics and data science training across the province. With a strong background in project management, event coordination, and stakeholder engagement, she strives to create impactful initiatives that foster collaboration and skill development. Her experience in organizing training programs, developing policies, and building supportive environments enables her to contribute to Ontario's leadership in the bioinformatics community.

2.0.3.2 Riya Roy (she/her)

Regional Coordinator, CBH Prairies Canadian Bioinformatics Hub, University of Manitoba Winnipeg, MB, Canada

— prairies@bioinformatics.ca

Riya is currently in the final year of her Ph.D. in Biomolecular Science, specializing in the molecular biology of non-coding RNAs. She holds a Bachelor's degree in Microbiology and a Master's degree in Bioinformatics from India. In 2025, Riya joined the Canadian Bioinformatics Hub as the Regional Coordinator for

the Prairie region. With a strong interest in science communication and public engagement, Riya is committed to advancing bioinformatics education and fostering stronger connections within both regional and national bioinformatics communities.

Chapter 3

1. Logging into AWS

3.0.1 Logging into the Amazon Cloud During the Workshop

- These instructions will **ONLY** be relevant in class, as the Cloud will not be accessible from home in advance of the class.
- On the cloud, we're going to use the default username: **ubuntu**

3.0.2 Logging in with ssh (Mac/Linux)

- Make sure the permissions on your private key are secure. Use `chmod` on your downloaded pem file:

```
chmod 600 CBW.pem
```

- To log in to the instance, use the `-i` command line argument to specify your private key:

```
ssh -i CBW.pem ubuntu@##.uhn-hpc.ca
```

(where `##` is your assigned student number.)

3.0.3 Logging in with OpenSSH client (Windows)

- Make sure OpenSSH client is installed on your machine. Settings > System > Optional features, then search for “OpenSSH” in your added features.

- open the “Command Prompt” or “Windows PowerShell” and run

```
C:\Windows\System32\OpenSSH\ssh.exe -i CBW.pem ubuntu@##.uhn-hpc.ca
```

3.1 File System Layout

When you log in, you’ll notice that you have two directories: **CourseData** and **workspace**.

- The **CourseData** directory will contain the files that you’ll need to complete your lab assignments.
- The **workspace** directory is where you will work on your labs. Files here will be accessible through your browser.

3.2 Workspace

- Everything created in your workspace on the cloud is also available by a web server on your cloud instance. Simply go to the following in your browser:

```
http://##.uhn-hpc.ca/
```

(where ## is your assigned student number.)

3.3 RStudio

- RStudio server is installed on your instance. It is accessible in your browser

```
http://##.uhn-hpc.ca:8080
```

(where ## is your assigned student number.) The username is **ubuntu**. We will give you the password in class.

3.4 Jupyter Notebook

- Jupyter Notebook is installed on your instance. It is accessible in your browser

`http://##.uhn-hpc.ca:8000`

(where ## is your assigned student number.) We will give you the password in class.

Chapter 4

2. Introduction to the command line

4.0.1 Exercise: Exploring the filesystem

1. Connect to your AWS instance
2. Type the `ls` command - what is the output?

Solution

```
$ ls
CourseData  R  workspace
```

The `ls` command lists the contents of a working directory.

3. Type the `pwd` command - what is the output?

Solution

```
$ pwd
/home/ubuntu
```

The `pwd` command shows the absolute *path to the working directory*.

Chapter 5

3. File manipulation

5.0.1 Exercise: Reading text files

Using the commands you just learned, explore the `.gff3` file in the Home directory.

1. Is the file a text file?

Solution

Yes. You can use `less`, `cat`, `head`, or `tail` and get human-readable info. Note that this doesn't have anything to do with its file extension.

2. How many lines does the file have?

Solution

```
$ wc -l GCF_009858895.2_ASM985889v3_genomic.gff
67 GCF_009858895.2_ASM985889v3_genomic.gff
```

There are 67 lines in this file.

3. Can you read the content of the file using `less`?

Solution

```
$ less GCF_009858895.2_ASM985889v3_genomic.gff
```

5.0.2 Exercise: Editing text files

Using the commands you just learned, create a file called `helloworld.txt` and edit it using `nano`.

1. Write “Hello world” into the file. Save the file and exit `nano`.
2. Create a subdirectory called “test”; move the `helloworld.txt` file into `test`.
3. Create a copy of the `helloworld.txt` file called `helloworld2.txt`

Solution

1. First, use the `nano` command to open a file called `helloworld.txt`

```
$ nano helloworld.txt
```

Inside the `nano` editor, write “Hello world”, then use the `^O` option to write the changes and `^X` to exit.

2. Create a subdirectory called “test”; move the `helloworld.txt` file into `test`.

Use the command `mkdir` to create this new directory. Then, use `mv` to move `helloworld.txt` into this directory.

```
$ mkdir test
$ mv helloworld.txt test/
```

3. Create a copy of the `helloworld.txt` file called `helloworld2.txt`.

Change the working directory using `cd`, then use the `cp` command to create the copy.

```
$ cd test
$ cp helloworld.txt helloworld2.txt
```

5.0.3 Reference material

Here are two cheat-sheets that can be useful to have as a reference for common UNIX/Linux commands:

- FOSSwire.com Unix/Linux Command Reference
- SUSO.org Unix/Linux Command Syntax and Reference

Chapter 6

4. Searching and sorting through files

6.0.1 Setup

We want to explore some data on our instances, but it is a compressed ZIP file, so we'll need to unzip it first. First move into your home directory, and then unzip the file named data.zip:

```
cd ~  
unzip data.zip  
cd data
```

Take a look at what's in your home directory after unzipping using ls:

```
ls
```

```
GCF_009858895.2_ASM985889v3_genomic.gff  __MACOSX  data  data.zip
```

Note that the original zipped file (data.zip) is still present, but there is now a folder (data) that you can enter. There is also a file called __MACOSX. This is because the folder was zipped using a Mac; you can safely ignore it here.

6.1 Data Exploration

We'll begin by looking at files are in Genbank gbff. This is a text file that describes the nucleotide sequence and annotation features on those nucleotide

sequences. First of all, we run the `ls` command to view the names of the files in the `ecoli_genomes` directory:

```
$ ls genomes
```

```
atlanta.gbff      braunschweig.gbff lab-strain.gbff   london.gbff      muenster.gbff
```

Let's go into that directory with `cd` and run an example command `wc london.gbff`:

```
$ cd genomes
$ wc london.gbff
```

```
212462  899913 13993868 london.gbff
```

`wc` is the 'word count' command: it counts the number of lines, words, and characters in files (from left to right, in that order).

If we run the command `wc *.gbff`, the `*` wildcard matches zero or more occurrences of any character, so the shell turns `*.gbff` into a list of all `gbff` files in the current directory:

```
$ wc *.gbff
```

```
217040  915629 14251456 atlanta.gbff
204528  837084 13332241 braunschweig.gbff
191859  685480 11851718 lab-strain.gbff
212462  899913 13993868 london.gbff
202036  823021 13167730 muenster.gbff
216788  883045 14082598 nevada.gbff
208514  849571 13555200 texas.gbff
1453227 5893743 94234811 total
```

Note that `wc *.gbff` also shows the total number of all lines in the last line of the output.

If we run `wc -l` instead of just `wc`, the output shows only the number of lines per file:

```
$ wc -l *.gbff
```

```
217040 atlanta.gbff
204528 braunschweig.gbff
191859 lab-strain.gbff
212462 london.gbff
202036 muenster.gbff
216788 nevada.gbff
208514 texas.gbff
1453227 total
```

Which of these files contains the fewest lines? It's an easy question to answer when there are only six files, but what if there were 6000? Our first step toward a solution is to run the command:

```
$ wc -l *.gbff > lengths.txt
```

The greater than symbol, `>`, tells the shell to redirect the command's output to a file instead of printing it to the screen. (This is why there is no screen output: everything that `wc` would have printed has gone into the file `lengths.txt` instead.) The shell will create the file if it doesn't exist. If the file exists, it will be silently overwritten, which may lead to data loss and thus requires some caution. `ls lengths.txt` confirms that the file exists:

```
$ ls lengths.txt
lengths.txt
```

We can now send the content of `lengths.txt` to the screen using `cat lengths.txt`. The `cat` command gets its name from 'concatenate' i.e. join together, and it prints the contents of files one after another. There's only one file in this case, so `cat` just shows us what it contains:

```
$ cat lengths.txt
```

```
217040 atlanta.gbff
204528 braunschweig.gbff
191859 lab-strain.gbff
212462 london.gbff
202036 muenster.gbff
216788 nevada.gbff
208514 texas.gbff
1453227 total
```

6.2 Sorting

The `sort` command rearranges the lines in a file in order. There are different methods of sorting - lexicographically (a-z1-9) or numerically. The default sort type is lexicographically, where numbers are treated one character at a time. Given a the file “numbers.txt” that looks like:

```
cd ../sorting
cat numbers.txt
```

```
10
2
19
22
6
```

If we run `sort` on this file:

```
sort numbers.txt
```

```
10
19
2
22
6
```

If we run `sort -n` on the same input - specifying that we want to sort numerically, we get this instead:

```
sort -n numbers.txt
```

```
2
6
10
19
22
```

Explain why `-n` has this effect.

Solution

The `-n` option specifies a numerical rather than an alphanumerical sort.

We will sort our `lengths.txt` file using the `-n` option to specify that the sort is numerical instead of alphanumerical. Note that running `sort` does not modify the file; instead, it sends the sorted result to the screen:

```
cd ../genomes
$ sort -n lengths.txt
```

```
191859 lab-strain.gbff
202036 muenster.gbff
204528 braunschweig.gbff
208514 texas.gbff
212462 london.gbff
216788 nevada.gbff
217040 atlanta.gbff
1453227 total
```

We can put the sorted list of lines in another temporary file called `sorted-lengths.txt` by putting `> sorted-lengths.txt` after the command, just as we used `> lengths.txt` to put the output of `wc` into `lengths.txt`. Once we've done that, we can run another command called `head` to get the first few lines in `sorted-lengths.txt`:

```
$ sort -n lengths.txt > sorted-lengths.txt
$ head -n 1 sorted-lengths.txt
```

```
191859 lab-strain.gbff
```

Using `-n 1` with `head` tells it that we only want the first line of the file; `-n 20` would get the first 20, and so on. Since `sorted-lengths.txt` contains the lengths of our files ordered from least to greatest, the output of `head` must be the file with the fewest lines.

6.2.1 What Does `>` Mean?

We have seen the use of `>`, but there is a similar operator `>>` which works slightly differently. We'll learn about the differences between these two operators by printing some strings. We can use the `echo` command to print strings e.g.

```
$ echo The echo command prints text
```

The `echo` command prints text

Now test the commands below to reveal the difference between the two operators:

```
$ echo hello > testfile01.txt
```

and:

```
$ echo hello >> testfile01.txt
```

6.2.2 Running Commands Together

If you think this is confusing, you're in good company: even once you understand what `wc`, `sort`, and `head` do, all those intermediate files make it hard to follow what's going on. We can make it easier to understand by running `sort` and `head` together:

```
$ sort -n lengths.txt | head -n 1
```

```
191859 lab-strain.gbff
```

The vertical bar, `|`, between the two commands is called a pipe. It tells the shell that we want to use the output of the command on the left as the input to the command on the right.

Nothing prevents us from chaining pipes consecutively. That is, we can for example send the output of `wc` directly to `sort`, and then the resulting output to `head`. Thus we first use a pipe to send the output of `wc` to `sort`:

```
$ wc -l *.gbff | sort -n
```

```
191859 lab-strain.gbff
202036 muenster.gbff
204528 braunschweig.gbff
208514 texas.gbff
212462 london.gbff
216788 nevada.gbff
217040 atlanta.gbff
1453227 total
```